MAT 2040: Linear Algebra

Assignment 4

- **Release date:** October 21, Monday.
- **Due date: October 31, Saturday.**
- Late submission is **Not** accepted.
- Please submit your answers as a PDF file with a name containing your student ID + ASS No. like "123456XXX ASS4.pdf".

1. Let $A = \begin{bmatrix} -3 & -2 & 0 \\ 0 & 2 & -6 \\ 6 & 3 & 3 \end{bmatrix}$ and $\boldsymbol{u} = \begin{bmatrix} 1 \\ 14 \\ -9 \end{bmatrix}$. Is $\boldsymbol{u}$ in Null(A)? Is $\boldsymbol{u}$ in Col(A)? Justify each answer.

2. Let $A = \begin{bmatrix} 3 & 2 & 1 & -5 \\ -9 & -4 & 1 & 7 \\ 9 & 2 & -5 & 1 \end{bmatrix}$

   (a) Give integers p and q such that Null(A) is a subspace of $\mathbb{R}^p$ and Col A is a subspace of $\mathbb{R}^q$.

   (b) Find a nonzero vector in Null(A) and a nonzero vector in Col(A).

3. Consider

$$A = \begin{bmatrix} 2 & -3 & 6 & 2 & 5 \\ -2 & 3 & -3 & -3 & -4 \\ 4 & -6 & 9 & 5 & 9 \\ -2 & 3 & 3 & -4 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & -3 & 6 & 2 & 5 \\ 0 & 0 & 3 & -1 & 1 \\ 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

   Assume that the matrix A is row equivalent to B. Without calculations, list rank(A) and dim(Null(A)). Then find bases for Col(A), Row(A), and Null(A).

4. Determine the dimensions of Null(A) and Col(A) for the following matrices:

   (a) $A = \begin{bmatrix} 1 & -6 & 9 & 0 & -2 \\ 0 & 1 & 2 & -4 & 5 \\ 0 & 0 & 0 & 5 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

(b) $A = \begin{bmatrix} 1 & 3 & -4 & 2 & -1 & 6 \\ 0 & 0 & 1 & -3 & 7 & 0 \\ 0 & 0 & 0 & 1 & 4 & -3 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

5. Find the dimension of the subspace spanned by the given vectors

(a) $\begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 9 \\ 4 \\ -2 \end{bmatrix}, \begin{bmatrix} -7 \\ -3 \\ 1 \end{bmatrix}$

(b) $\begin{bmatrix} 1 \\ -2 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 4 \\ 1 \end{bmatrix}, \begin{bmatrix} -8 \\ 6 \\ 5 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ 7 \end{bmatrix}$

6. Find the vector $\boldsymbol{x}$ determined by the given coordinate vector $[\boldsymbol{x}]_{\mathcal{B}}$ and the given basis $\mathcal{B}$:

(a) $\mathcal{B} = \left\{ \begin{bmatrix} 3 \\ -5 \end{bmatrix}, \begin{bmatrix} -4 \\ 6 \end{bmatrix} \right\}, \quad [\boldsymbol{x}]_{\mathcal{B}} = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$

(b) $\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ -4 \\ 3 \end{bmatrix}, \begin{bmatrix} 5 \\ 2 \\ -2 \end{bmatrix}, \begin{bmatrix} 4 \\ -7 \\ 0 \end{bmatrix} \right\}, \quad [\boldsymbol{x}]_{\mathcal{B}} = \begin{bmatrix} 3 \\ 0 \\ -1 \end{bmatrix}$

(c) $\mathcal{B} = \left\{ \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ -5 \\ 2 \end{bmatrix}, \begin{bmatrix} 4 \\ -7 \\ 3 \end{bmatrix} \right\}, \quad [\boldsymbol{x}]_{\mathcal{B}} = \begin{bmatrix} -4 \\ 8 \\ 7 \end{bmatrix}$

7. Let $V = \text{Span}\{\boldsymbol{v}_1, \boldsymbol{v}_2\}$ and $\mathcal{B} = \{\boldsymbol{v}_1, \boldsymbol{v}_2\}$. Show that $\boldsymbol{a}$ is in V, and find the $\mathcal{B}$-coordinate vector of $\boldsymbol{a}$, when

$$\boldsymbol{v}_1 = \begin{bmatrix} 11 \\ -5 \\ 10 \\ 7 \end{bmatrix}, \quad \boldsymbol{v}_2 = \begin{bmatrix} 14 \\ -8 \\ 13 \\ 10 \end{bmatrix}, \quad \boldsymbol{a} = \begin{bmatrix} 19 \\ -13 \\ 18 \\ 15 \end{bmatrix}$$

8. Find the transition matrix from $\mathcal{B}$ to the standard basis in $\mathcal{R}^n$:

(a) $\mathcal{B} = \left\{ \begin{bmatrix} 2 \\ -9 \end{bmatrix}, \begin{bmatrix} 1 \\ 8 \end{bmatrix} \right\}$

(b) $\mathcal{B} = \left\{ \begin{bmatrix} 3 \\ -1 \\ 4 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ -5 \end{bmatrix}, \begin{bmatrix} 8 \\ -2 \\ 7 \end{bmatrix} \right\}$

9. True or false (with a counterexample if false):

(a) If Col A contains only the zero vector, then A is the zero matrix.

(b) The column space of $2A$ equals the column space of A.

(c) The column space of $A - I$ equals the column space of A.

(d) $A \in \mathbb{R}^{m \times n}$ has no more than n pivot columns.

(e) A and A^T have the same null space.

10. The set $\mathcal{B} = \{1 + t^2, t + t^2, 1 + 2t + t^2\}$ is a basis for $\mathbb{P}_2$. Find the coordinate vector of $p(t) = 1 + 4t + 7t^2$ relative to $\mathcal{B}$.

11. The first four Hermite polynomials are 1, $2t$, $-2 + 4t^2$, and $-12t + 8t^3$. There polynomials arise naturally in the study of certain important differential equations in mathematical physics. Show that the first four Hermite polynomials form a basis of $\mathbb{P}^3$.

12. Suppose that $A \in \mathbb{R}^{m \times n}$

    (a) Show that if $A\boldsymbol{x} = \boldsymbol{b}$ is consistent for all $\boldsymbol{b} \in \mathbb{R}^m$, then $n(A) = n - m$.

    (b) Null(A)=Null($A^T A$)

    (c) rank(A)=rank($A^T A$)

13. If $A \in \mathbb{R}^{3 \times 8}$ has rank 2, find dim(Null(A)), dim(Row(A)) and rank(A^T).

14. Verify that rank($\boldsymbol{u}\boldsymbol{v}^T$) = 1 if $\boldsymbol{u} = \begin{bmatrix} 2 \\ -3 \\ 5 \end{bmatrix}$ and $\boldsymbol{v} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$.

15. Prove the following useful theorem: If W_1 and W_2 are two subspaces of a finite dimensional vector space V, and $W_1 \cap W_2 = \{0\}$, then $W_1 + W_2 = \{w | w = w_1 + w_2, w_1 \in W_1 \text{ and } w_2 \in W_2\}$ is a finite dimensional vector space:

$$\dim(W_1 + W_2) = \dim(W_1) + \dim(W_2)$$